

A p -Lorentz-core reverse inclusion for the endpoint complex noncommutative martingale Hardy–BMO problem

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Status

This is an unconditional partial result for the endpoint complex interpolation problem in Randrianantoanina’s paper [1]. It proves that the p -Lorentz Hardy core $\mathbf{h}_{r,p}^c(\mathcal{M})$ lies in the endpoint complex interpolation space. It does not prove that all of $\mathbf{h}_r^c(\mathcal{M})$ lies there.

Source problem

The source paper proves the finite-index complex interpolation theorem

$$[\mathbf{h}_p^c(\mathcal{M}), \mathbf{h}_q^c(\mathcal{M})]_\theta = \mathbf{h}_r^c(\mathcal{M}), \quad \frac{1}{r} = \frac{1-\theta}{p} + \frac{\theta}{q}, \quad 0 < p, q < \infty.$$

It then asks whether the corresponding endpoint identity remains valid when $0 < p < 1$ and the endpoint couple $(\mathbf{h}_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M}))$ is used.

From combining (3.12) and (3.13), we obtain the desired equivalence. $\square$

When $0 < p < 1$, we do not know if the corresponding statement to Theorem 3.16 remains valid if the interpolation couple $(\mathbf{h}_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M}))$ is used. Since reiteration theorem is not available for complex interpolations of quasi-Banach spaces, in general, this consideration is independent of Theorem 3.16. We leave this as an open problem.

The same method of proofs can be applied to conditioned spaces and spaces of adapted se-

The full desired endpoint identity is

$$[\mathbf{h}_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta = \mathbf{h}_r^c(\mathcal{M}), \quad \frac{1}{r} = \frac{1-\theta}{p}.$$

Idea of the proof

The Cuculescu decomposition in [2] proves the real endpoint scale

$$(\mathbf{h}_p^c, \mathbf{h}_\infty^c)_{\theta, \gamma} = \mathbf{h}_{r, \gamma}^c, \quad \frac{1}{r} = \frac{1-\theta}{p}.$$

Choosing $\gamma = p$ gives $\mathbf{h}_{r,p}^c = (\mathbf{h}_p^c, \mathbf{h}_\infty^c)_{\theta, p}$. The couple $(\mathbf{h}_p^c, \mathbf{h}_\infty^c)$ is naturally p -Banach: the $\mathbf{h}_p^c$ quasi-norm is p -subadditive and the $\mathbf{h}_\infty^c$ norm is also p -subadditive because $0 < p < 1$. For a p -Banach

couple, the real space with summability index p embeds into the complex interpolation space by the usual analytic series construction. Thus

$$\mathbf{h}_{r,p}^c \hookrightarrow [\mathbf{h}_p^c, \mathbf{h}_\infty^c]_\theta.$$

The endpoint inclusion $\mathbf{h}_\infty^c \hookrightarrow \mathbf{bmo}^c$ and complex functoriality pass this into the $\mathbf{bmo}^c$ -endpoint couple. The already packaged one-sided Wolff argument gives the opposite-side bound into $\mathbf{h}_r^c$.

Lemma 1 (real (θ, ρ) embeds into complex interpolation). *Let (A_0, A_1) be a compatible couple whose endpoint quasi-norms are ρ -subadditive for some $0 < \rho \leq 1$, and let $0 < \theta < 1$. Then*

$$(A_0, A_1)_{\theta, \rho} \hookrightarrow [A_0, A_1]_\theta$$

continuously.

Proof. By the discrete J -method characterization of $(A_0, A_1)_{\theta, \rho}$, every x in this real interpolation space admits, up to an arbitrarily small loss in the norm, a representation

$$x = \sum_{n \in \mathbb{Z}} x_n, \quad x_n \in A_0 \cap A_1,$$

with convergence in $A_0 + A_1$ and

$$\sum_{n \in \mathbb{Z}} (2^{-n\theta} J(2^n, x_n; A_0, A_1))^\rho < \infty, \quad J(t, u) = \max\{\|u\|_{A_0}, t\|u\|_{A_1}\}.$$

For finite partial sums define

$$F_N(z) = \sum_{|n| \leq N} 2^{n(z-\theta)} x_n.$$

Then $F_N(\theta) = \sum_{|n| \leq N} x_n$. On the boundary $\Re z = 0$,

$$\|2^{n(z-\theta)} x_n\|_{A_0} = 2^{-n\theta} \|x_n\|_{A_0} \leq 2^{-n\theta} J(2^n, x_n),$$

and on the boundary $\Re z = 1$,

$$\|2^{n(z-\theta)} x_n\|_{A_1} = 2^{n(1-\theta)} \|x_n\|_{A_1} \leq 2^{-n\theta} J(2^n, x_n).$$

The ρ -summability above and the endpoint ρ -triangle inequalities make the series converge uniformly on both boundary lines and uniformly on compact subsets of the strip in $A_0 + A_1$. Passing to the limit produces an admissible analytic function F with $F(\theta) = x$, and the complex norm is bounded by a constant times the real (θ, ρ) -norm. $\square$

Lemma 2. *There is a continuous embedding*

$$\mathbf{h}_\infty^c(\mathcal{M}) \hookrightarrow \mathbf{bmo}^c(\mathcal{M}).$$

Proof. For finite martingales this follows because bounded conditioned square function controls every conditioned BMO tail:

$$\mathcal{E}_n(|x_m - x_n|^2) = \mathcal{E}_n\left(\sum_{k=n+1}^m |dx_k|^2\right) = \mathcal{E}_n\left(\sum_{k=n+1}^m \mathcal{E}_{k-1}(|dx_k|^2)\right) \leq \|s_c(x)^2\|_\infty \mathbf{1}.$$

The first-coordinate term is controlled by the usual $\mathbf{h}_\infty^c$ norm. Density and completion give the endpoint embedding. $\square$

Theorem 1 (p -Lorentz-core endpoint inclusion). *Let $0 < p < 1$, $0 < \theta < 1$, and put*

$$\frac{1}{r} = \frac{1 - \theta}{p}.$$

Assume $r > 1$. Then

$$\mathbf{h}_{r,p}^c(\mathcal{M}) \hookrightarrow [\mathbf{h}_p^c(\mathcal{M}), \mathbf{h}_\infty^c(\mathcal{M})]_\theta \hookrightarrow [\mathbf{h}_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta \hookrightarrow \mathbf{h}_r^c(\mathcal{M}).$$

Proof. By Randrianantoanina's endpoint real interpolation theorem [2, Proposition 2.8],

$$(\mathbf{h}_p^c(\mathcal{M}), \mathbf{h}_\infty^c(\mathcal{M}))_{\theta, \gamma} = \mathbf{h}_{r, \gamma}^c(\mathcal{M})$$

for every $0 < \gamma \leq \infty$. Taking $\gamma = p$ gives

$$\mathbf{h}_{r,p}^c(\mathcal{M}) = (\mathbf{h}_p^c(\mathcal{M}), \mathbf{h}_\infty^c(\mathcal{M}))_{\theta, p}.$$

The preceding real-to-complex lemma, applied with $\rho = p$, yields

$$\mathbf{h}_{r,p}^c(\mathcal{M}) \hookrightarrow [\mathbf{h}_p^c(\mathcal{M}), \mathbf{h}_\infty^c(\mathcal{M})]_\theta.$$

Since $\mathbf{h}_\infty^c(\mathcal{M}) \hookrightarrow \mathbf{bmo}^c(\mathcal{M})$, functoriality of the complex method gives

$$[\mathbf{h}_p^c(\mathcal{M}), \mathbf{h}_\infty^c(\mathcal{M})]_\theta \hookrightarrow [\mathbf{h}_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta.$$

It remains only to recall why the final inclusion holds in the range $r > 1$. Choose $\eta \in (0, 1)$ so that

$$1 = \frac{1 - \eta}{p} + \frac{\eta}{r}.$$

The finite-index complex theorem [1, Theorem 3.16] gives

$$[\mathbf{h}_p^c(\mathcal{M}), \mathbf{h}_r^c(\mathcal{M})]_\eta = \mathbf{h}_1^c(\mathcal{M}).$$

With $\phi = 1 - 1/r$, the Banach endpoint theorem of Bekjan–Chen–Perrin–Yin [3, Theorem 4.4], after reversing the couple, gives

$$[\mathbf{h}_1^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\phi = \mathbf{h}_r^c(\mathcal{M}).$$

Wolff's inclusion lemma, in the inclusion form used in [1, Corollary 3.19], therefore implies

$$[\mathbf{h}_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta \hookrightarrow \mathbf{h}_r^c(\mathcal{M}),$$

with the usual parameter identity $\theta = \phi/(1 - \eta + \eta\phi)$. This completes the chain. $\square$

Verification notes

The proof is symbolic and uses no computation. The new inclusion is unconditional because every input is already a theorem: the 2022 real endpoint interpolation theorem, the elementary p -Banach (θ, p) -real-to-complex embedding, the endpoint $\mathbf{h}_\infty^c \subset \mathbf{bmo}^c$ estimate, and the previously used one-sided Wolff inclusion.

Limitations

This does not prove the full reverse inclusion

$$h_r^c(\mathcal{M}) \hookrightarrow [h_p^c(\mathcal{M}), bmo^c(\mathcal{M})]_\theta.$$

It only proves it on the Lorentz subspace $h_{r,p}^c(\mathcal{M})$. In general $L_{r,p} \subsetneq L_r$, so this should be treated as a proper partial result. Since $p < 1$, this is a smaller core than $h_{r,1}^c$. The missing part is the passage from the quasi-Banach summability index p to the natural index r , exactly the place where a stronger analytic family or endpoint complex theorem would be needed.

Novelty check

Local indexes were searched for arXiv:2108.06341, h_p^c , bmo^c , h_∞^c , $h_{r,p}^c$, Lorentz core, and endpoint complex interpolation. The existing run packets contained the one-sided $r > 1$ inclusion and a conditional h_∞^c -endpoint reduction, but no durable unconditional p -Lorentz-core reverse inclusion packet. External checks in the same loop found the source paper [1] and the real endpoint paper [2]; neither states the above p -Lorentz-core complex consequence as the answer to the source endpoint problem.

Human review recommendation

Review as an unconditional partial theorem. The main checks are the p -Banach real-to-complex embedding $(A_0, A_1)_{\theta,p} \hookrightarrow [A_0, A_1]_\theta$, the use of [2] with Lorentz parameter p , and the endpoint embedding $h_\infty^c \hookrightarrow bmo^c$. The packet should not be read as claiming the full reverse inclusion for h_r^c .

References

- [1] N. Randrianantoanina, *Interpolation between noncommutative martingale Hardy and BMO spaces: the case $0 < p < 1$* , arXiv:2108.06341.
- [2] N. Randrianantoanina, *P. Jones' interpolation theorem for noncommutative martingale Hardy spaces*, arXiv:2212.08714.
- [3] T. N. Bekjan, Z. Chen, M. Perrin, and Z. Yin, *Atomic decomposition and interpolation for Hardy spaces of noncommutative martingales*, arXiv:0906.4437.
- [4] T. Wolff, *A note on interpolation spaces*, Harmonic Analysis (Minneapolis, Minn., 1981), Lecture Notes in Math. 908, Springer, 1982, 199–204.